23- S1- Q3

$$Q: \ 1000 \begin{cases} 600 & : 1 \\ 400 & : 2 \end{cases}$$

$$m_1 = \begin{bmatrix} 0.023 \\ 0.061 \end{bmatrix} \qquad m_2 = \begin{bmatrix} 1.45 \\ 1.98 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 615.87 & 55.83 \\ 55.83 & 539.64 \end{bmatrix} \qquad S_2 = \begin{bmatrix} 426.49 & -1.77 \\ -1.77 & 387.20 \end{bmatrix}$$

(a) DLA

$$X_1 = \begin{bmatrix} 1.52 \\ -1.61 \end{bmatrix}$$

(b) $$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]}$$

$$NBC \rightarrow X_2 = \begin{bmatrix} 0.51 \\ 2.25 \end{bmatrix}$$

Solution ① LDA

For two classes the Fisher linear discriminant is

$$w = S_w^{-1}(m_1 - m_2)$$

$$w_0 = -\frac{1}{2}w^T(m_1 + m_2)$$

where $S_w = S_1 + S_2$

the discriminant function is

$$g(x) = w^T x + w_0$$

② $S_w = S_1 + S_2$

$$= \begin{bmatrix} 1042.36 & 54.06 \\ 54.06 & 926.84 \end{bmatrix}$$

$$m_1 - m_2 = \begin{bmatrix} 1.473 \\ 2.041 \end{bmatrix}$$

$$w = S_w^{-1}(m_1 - m_2)$$

$$= \begin{bmatrix} 0.001303 \\ 0.002126 \end{bmatrix}$$

$$w_0 = -\frac{1}{2}w^T(m_1 + m_2)$$

$$= -0.003128$$

③ $g(x) = \begin{bmatrix} 0.001303 & 0.002126 \end{bmatrix} \begin{bmatrix} 1.52 \\ -1.61 \end{bmatrix} - 0.003128$

$$= -0.0045703 < 0$$

④ For class 1, m_1

$$g(x) = \begin{bmatrix} 0.001303 & 0.002126 \end{bmatrix} \begin{bmatrix} 0.023 \\ 0.061 \end{bmatrix} - 0.003128$$

$$= -0.002968 < 0$$

For class 2, m_2

$$g(x) = \begin{bmatrix} 0.001303 & 0.002126 \end{bmatrix} \begin{bmatrix} 1.45 \\ 1.98 \end{bmatrix} - 0.003128$$

$$= 0.002971$$

If $g(x) < 0$, classifying x to class 1

If $g(x) > 0$, classifying x to class 2

If $g(x) = 0$, x on decision boundary

⑤ $g(x_1) < 0$

So we predict class 1 for x_1

(b) ① Gaussian Naïve Bayes

$$p(w_j|x) = \frac{p(x|w_j)\,p(w_j)}{p(x)} = \frac{\prod_{i=1}^{d} p(x_i|w_j)\,p(w_j)}{p(x)}$$

$$p(x|w_j) = \frac{1}{\sqrt{2\pi}\,\sigma_j} e^{-\frac{1}{2}\left(\frac{x-\mu_j}{\sigma_j}\right)^2}$$

The discriminant function is

$$g_i(x) = \prod_{i=1}^{d} p(x_i|w_j)\,p(w_j)$$

for $x_2 = \begin{bmatrix} 0.51 \\ 2.25 \end{bmatrix}$

we denote $x_1 = 0.51$, $x_2 = 2.25$

$$p(w_1) = \frac{600}{1000} = 0.6$$

$$p(w_2) = \frac{400}{1000} = 0.4$$

② calculate Σ_1 and Σ_2

$$\Sigma_1 = \frac{1}{n_1} S_1 = \frac{1}{600} S_1 = \begin{bmatrix} 1.02645 & 0.09305 \\ 0.09305 & 0.8994 \end{bmatrix}$$

$$\Sigma_2 = \frac{1}{n_2} S_2 = \frac{1}{400} S_2 = \begin{bmatrix} 1.066225 & -0.004425 \\ -0.004425 & 0.968 \end{bmatrix}$$

③

we denote σ_{ij}^2 as feature i variance of class w_j

so, $\sigma_{11}^2 = 1.02645$ $\quad \sigma_{21}^2 = 0.8994$

$\sigma_{12}^2 = 1.066225$ $\quad \sigma_{22}^2 = 0.968$

we denote μ_{ij} as feature i mean of class w_j

so, $\mu_{11} = 0.023$ $\quad \mu_{21} = 0.061$

$\mu_{12} = 1.45$ $\quad \mu_{22} = 1.98$

② $$p(x_1|w_1) = \frac{1}{\sqrt{2\pi}\,\sigma_{11}} e^{-\frac{1}{2}\left(\frac{x_1-\mu_{11}}{\sigma_{11}}\right)^2}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1.02645}} e^{-\frac{1}{2}\left(\frac{0.51-0.023}{\sqrt{1.02645}}\right)^2}$$

$$= 0.3508$$

$$p(x_2|w_1) = \frac{1}{\sqrt{2\pi}\,\sigma_{21}} e^{-\frac{1}{2}\left(\frac{x_2-\mu_{21}}{\sigma_{21}}\right)^2}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{0.8994}} e^{-\frac{1}{2}\left(\frac{2.25-0.061}{\sqrt{0.8994}}\right)^2}$$

$$= 0.02948$$

$$p(x_1|w_2) = \frac{1}{\sqrt{2\pi}\,\sigma_{12}} e^{-\frac{1}{2}\left(\frac{x_1-\mu_{12}}{\sigma_{12}}\right)^2}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1.06625}} e^{-\frac{1}{2}\left(\frac{0.51-1.45}{\sqrt{1.06625}}\right)^2}$$

$$= 0.2553$$

$$p(x_2|w_2) = \frac{1}{\sqrt{2\pi}\,\sigma_{22}} e^{-\frac{1}{2}\left(\frac{x_1-\mu_1}{\sigma_{22}}\right)^2}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{0.968}} e^{-\frac{1}{2}\left(\frac{2.25-1.98}{\sqrt{0.968}}\right)^2}$$

$= 0.3906$

$g_1(x_2) = p(x_1|w_1)\, p(x_2|w_1)\, p(w_1)$

$= 0.3508 \times 0.02948 \times 0.6$

$= 0.006205$

$g_2(x_2) = p(x_1|w_2)\, p(x_2|w_2)\, p(w_2)$

$= 0.2553 \times 0.3906 \times 0.4$

$= 0.03989$

$g_2(x_2) > g_1(x_2)$

So, we classify x_2 to class 2